Puberty and emotions study exploratory model output

Prior predictive checks

```
=== Mediator equation yrep ===

Range: -24.37 22.3

Mean: 0

Median: 0

Standard deviation: 2.19

Quantiles (1%, 5%, 50%, 95%, 99%):

   1%    5%   50%   95%   99%
-5.89 -3.41  0.00  3.40  5.88


=== Outcome equation yrep ===

Range: -25.97 31.79

Mean: 0

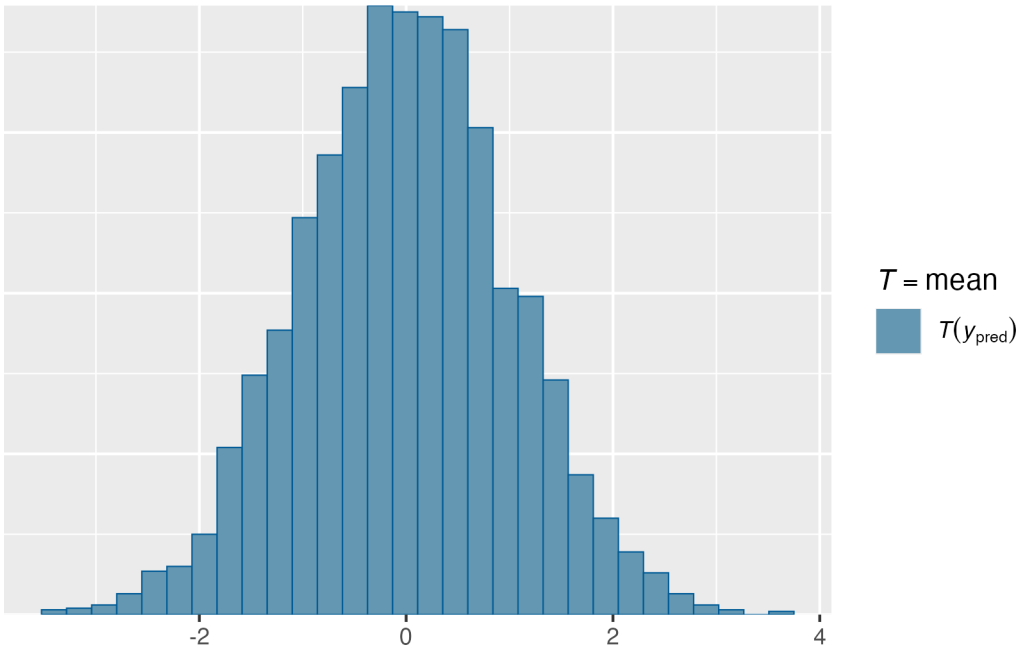
Median: -0.01

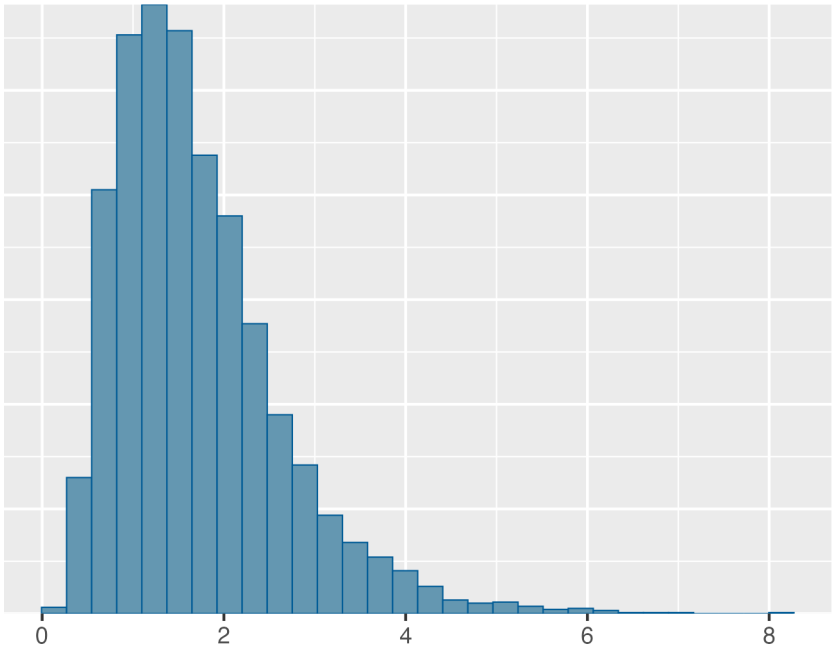
Standard deviation: 2.28
```

```
Quantiles (1%, 5%, 50%, 95%, 99%):

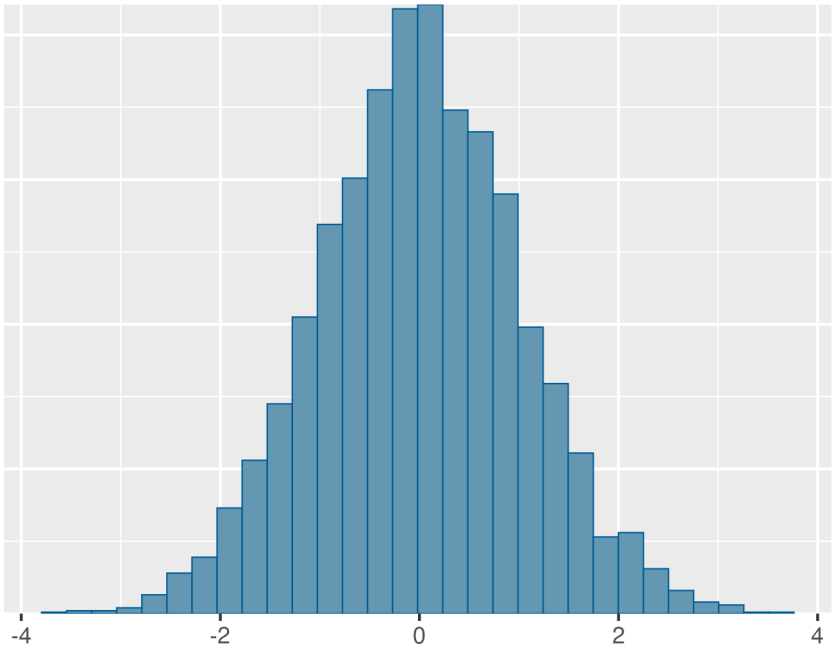
   1%    5%   50%   95%   99%
-5.91 -3.58 -0.01  3.62  6.01

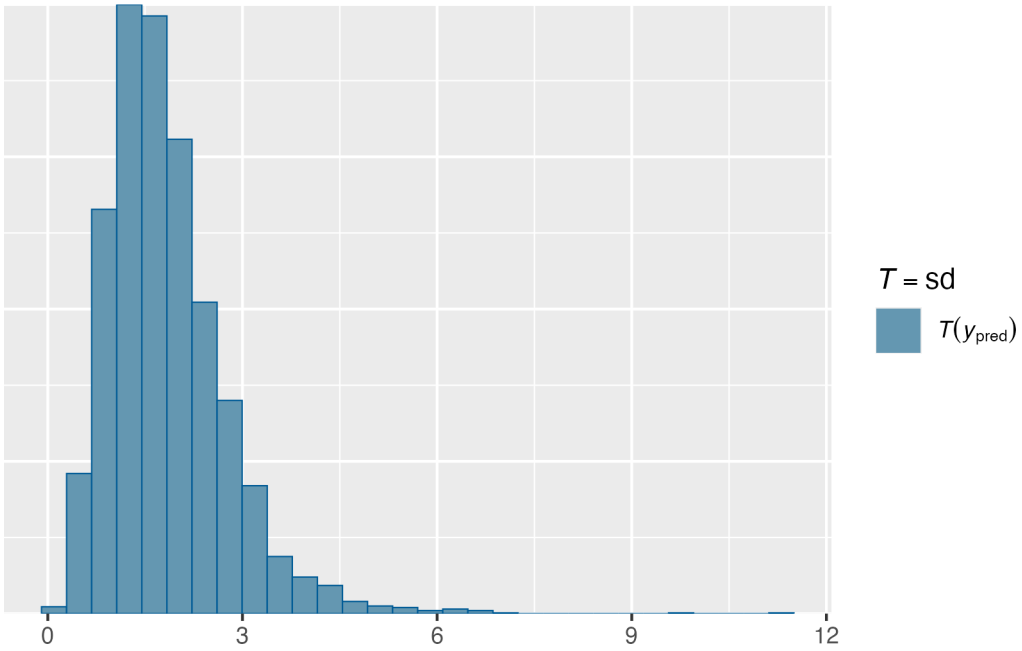
=== Yrep mediator mean and SD ===
```






```
=== Yrep outcome mean and SD ===
```






```
=== prior summary: fit_med_prior ===
```

prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
normal(0, 1)	b			erstotals						user
normal(0, 1)	b	adhd_statusY		erstotals						(vectorized)
normal(0, 1)	b	pubertal_timing_s		erstotals						(vectorized)
normal(0, 1)	b	pubertal_timing_s:adhd_statusY		erstotals						(vectorized)
normal(0, 1)	b			smfqtotals						user
normal(0, 1)	b	adhd_statusY		smfqtotals						
normal(0, 1)	b	ers_total_s		smfqtotals						
normal(0, 1)	b	pubertal_timing_s		smfqtotals						
normal(0, 1)	Intercept			erstotals						
normal(0, 1)	Intercept			smfqtotals						
lkj_corr_cholesky(5)	Lrescor									
exponential(1)	sigma			erstotals						
exponential(1)	sigma			smfqtotals						

```
                    (vectorized)
                    (vectorized)
                    (vectorized)
                            user
                            user
                            user
             0              user
             0              user


=== model summary: fit_med_prior ===

 Family: MV(gaussian, gaussian)
  Links: mu = identity
         mu = identity
Formula: ers_total_s ~ pubertal_timing_s * adhd_status
         smfq_total_s ~ pubertal_timing_s + adhd_status + ers_total_s
   Data: analytic_dfs[[1]] (Number of observations: 124)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                                     Estimate Est.Error l-95% CI u-95% CI
erstotals_Intercept                      0.00      1.22    -2.41     2.38
smfqtotals_Intercept                    -0.00      1.20    -2.34     2.39
erstotals_pubertal_timing_s             -0.01      1.00    -1.98     1.91
erstotals_adhd_statusY                  -0.01      1.01    -1.97     1.94
erstotals_pubertal_timing_s:adhd_statusY 0.00      1.00    -1.92     1.95
smfqtotals_pubertal_timing_s            -0.01      0.99    -1.94     1.92
smfqtotals_adhd_statusY                  0.01      0.99    -1.95     1.95
smfqtotals_ers_total_s                  -0.01      1.01    -2.02     1.97
                                     Rhat Bulk_ESS Tail_ESS
erstotals_Intercept                  1.00     6293     2934
smfqtotals_Intercept                 1.00     6602     2980
erstotals_pubertal_timing_s          1.00     6813     2959
erstotals_adhd_statusY               1.00     5709     3424
erstotals_pubertal_timing_s:adhd_statusY 1.00     6739     3152
smfqtotals_pubertal_timing_s         1.00     6642     2769
smfqtotals_adhd_statusY              1.00     6964     2771
smfqtotals_ers_total_s               1.00     7463     3270

Further Distributional Parameters:
```

```
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma_erstotals      1.00      0.98     0.02     3.61 1.00     5120     2481
sigma_smfqtotals     0.99      0.98     0.03     3.64 1.00     5263     2298

Residual Correlations:
                            Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
rescor(erstotals,smfqtotals)    -0.00      0.29    -0.55     0.56 1.00     7146
                            Tail_ESS
rescor(erstotals,smfqtotals)     2839

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Moderated Mediation output

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# A tibble: 6 x 6
  Effect                       Median `Lower 95% CI` `Upper 95% CI`    PD  ROPE
  <chr>                         <dbl>          <dbl>          <dbl> <dbl> <dbl>
1 Indirect (non-ADHD)            0.19           0.01           0.4   99.0  18.4
2 Indirect (ADHD)               -0.01          -0.16           0.12  58.0  85.0
3 Index of Moderated Mediation  -0.2           -0.46           0.01  97.8  20.2
4 Direct                         0.02          -0.13           0.17  61.9  79.1
5 Total (non-ADHD)               0.22           0.01           0.44  97.9  13.9
6 Total (ADHD)                   0.01          -0.2            0.21  54.4  66.4
```

Moderated Mediation diagnostics

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.
```

```
# A tibble: 21 x 4
   variable                                rhat ess_bulk ess_tail
   <chr>                                  <dbl>    <dbl>    <dbl>
 1 b_erstotals_Intercept                   1.00  132587.   91859.
 2 b_smfqtotals_Intercept                  1.00   45445.   61401.
 3 b_erstotals_pubertal_timing_s           1.00   80510.   91061.
```

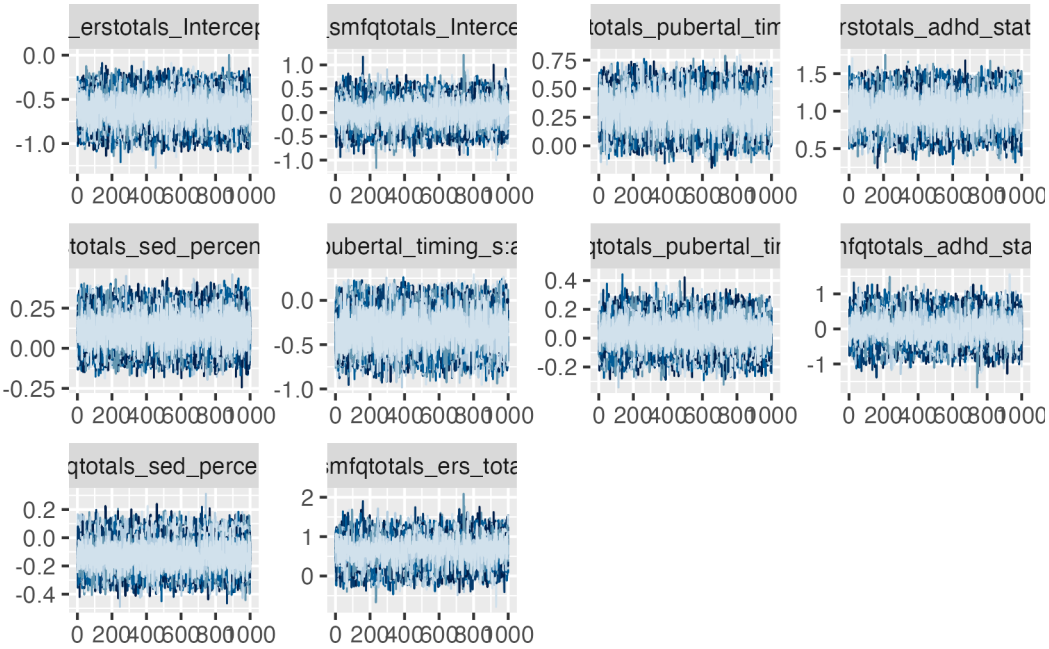
```
 4 b_erstotals_adhd_statusY                      1.00  134894.   92236.
 5 b_erstotals_sed_percentile_s                  1.00   27968.   91747.
 6 b_erstotals_pubertal_timing_s:adhd_statusY    1.00   77987.   89274.
 7 b_smfqtotals_pubertal_timing_s                1.00   50372.   82346.
 8 b_smfqtotals_adhd_statusY                     1.00   41227.   57693.
 9 b_smfqtotals_sed_percentile_s                 1.05    1297.    3804.
10 b_smfqtotals_ers_total_s                      1.00   34954.   46922.
# i 11 more rows


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



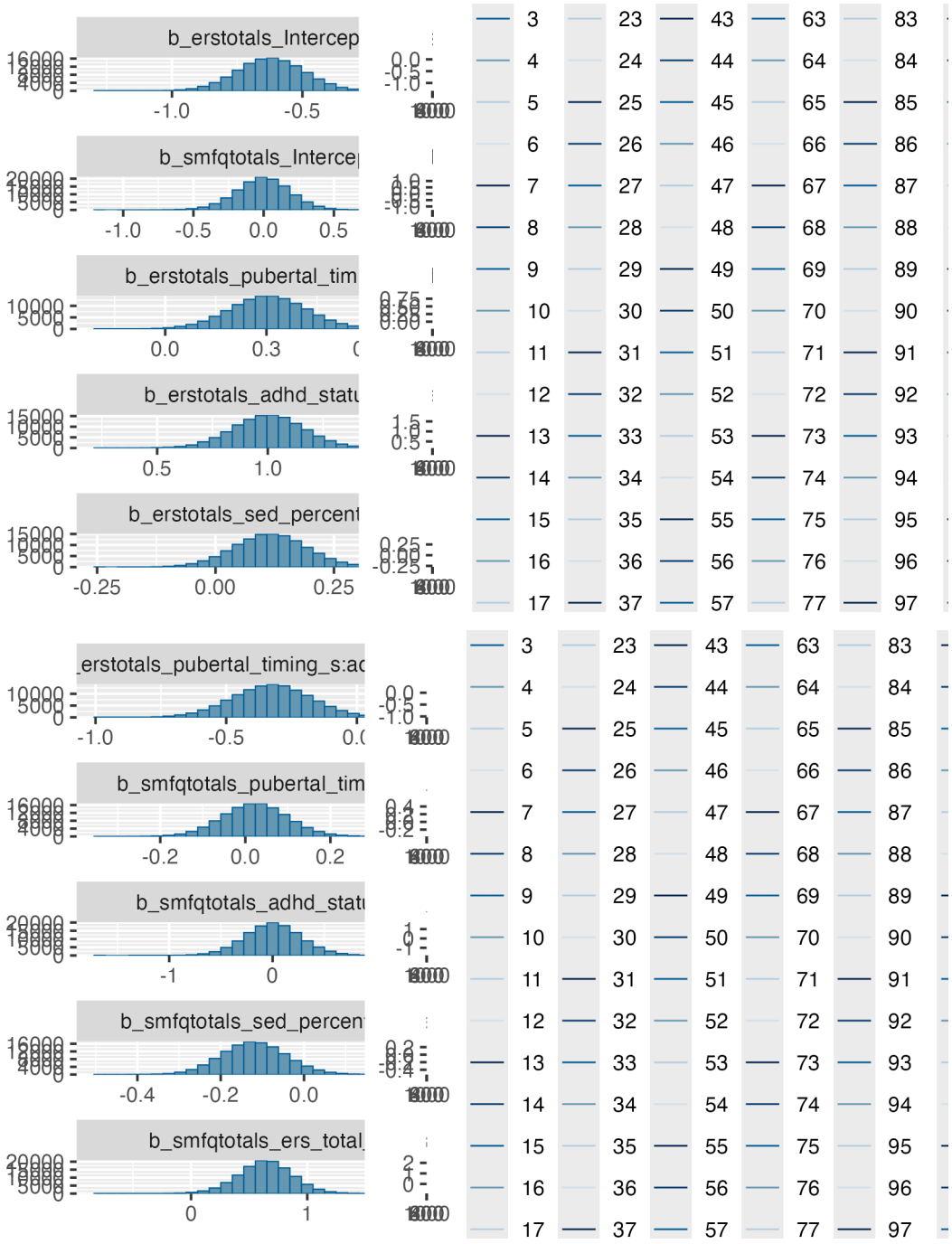

```
=== Prior Summary ===
```

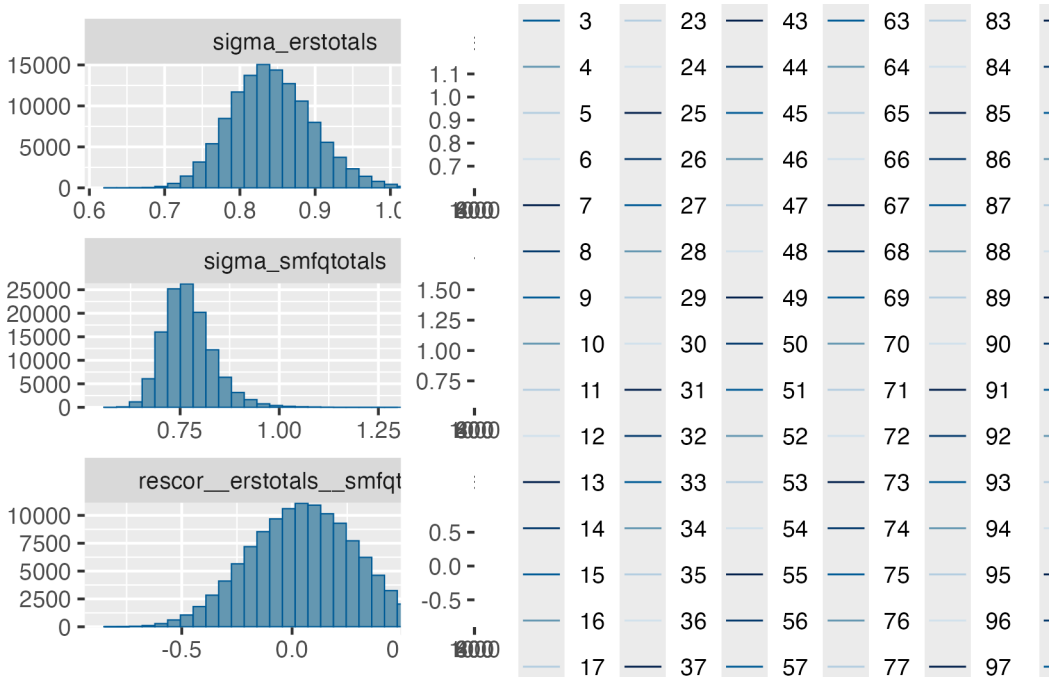
Priors specified for the model.

prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
normal(0, 1)	b			erstotals						user
normal(0, 1)	b	adhd_statusY		erstotals						(vectorized)
normal(0, 1)	b	pubertal_timing_s		erstotals						(vectorized)
normal(0, 1)	b	pubertal_timing_s:adhd_statusY		erstotals						(vectorized)
normal(0, 1)	b	sed_percentile_s		erstotals						(vectorized)
normal(0, 1)	b			smfqtotals						user
normal(0, 1)	b	adhd_statusY		smfqtotals						(vectorized)
normal(0, 1)	b	ers_total_s		smfqtotals						(vectorized)
normal(0, 1)	b	pubertal_timing_s		smfqtotals						(vectorized)
normal(0, 1)	b	sed_percentile_s		smfqtotals						(vectorized)
normal(0, 1)	Intercept			erstotals						user
normal(0, 1)	Intercept			smfqtotals						(vectorized)
lkj_corr_cholesky(5)	Lrescor									(vectorized)
exponential(1)	sigma			erstotals						user
exponential(1)	sigma			smfqtotals						user
							0			user
							0			user

=== Posterior Predictive Checks ===

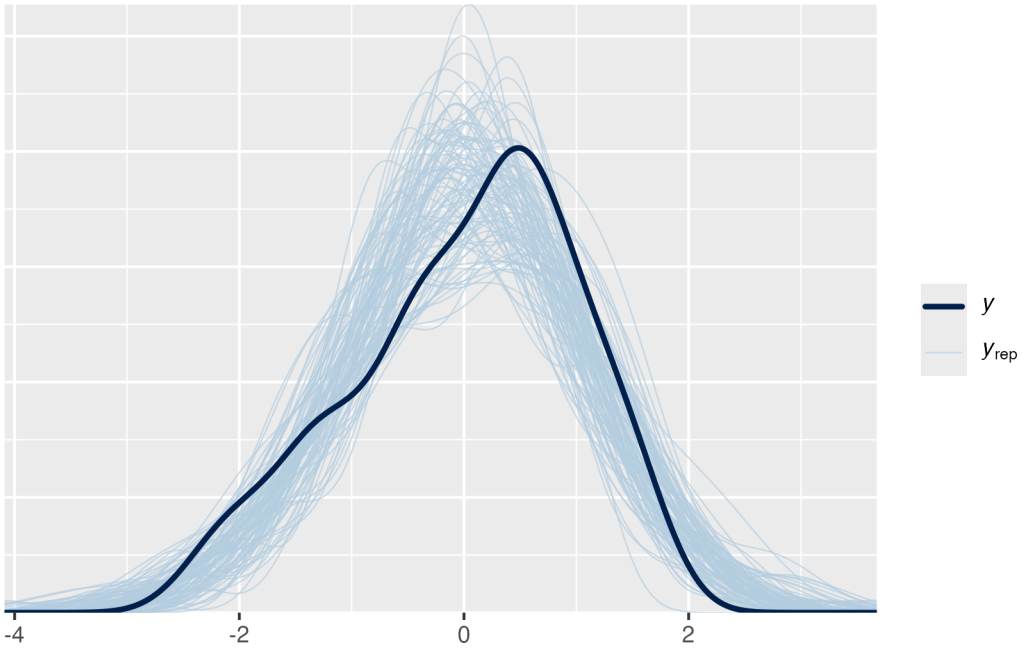
Density plots for each parameter.


b_erstotals_Intercep
b_smfqtotals_Interce
b_erstotals_pubertal_tim
b_erstotals_adhd_statu
b_erstotals_sed_percent
_erstotals_pubertal_timing_s:ad
b_smfqtotals_pubertal_tim
b_smfqtotals_adhd_statu
b_smfqtotals_sed_percen
b_smfqtotals_ers_total
3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
23 24 25 26 27 28 29 30 31 32 33 34 35 36 37
43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
63 64 65 66 67 68 69 70 71 72 73 74 75 76 77
83 84 85 86 87 88 89 90 91 92 93 94 95 96 97

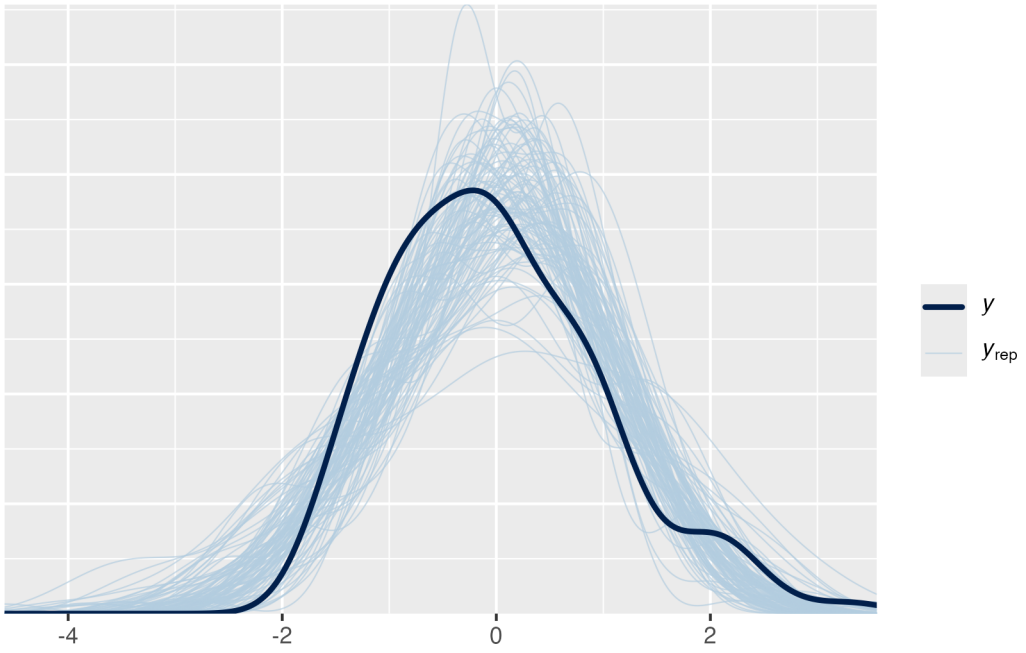



```
=== PPC: Density Overlay (Mediator: ers_total_s) ===

y_rep lines should closely follow the observed y distribution.
```

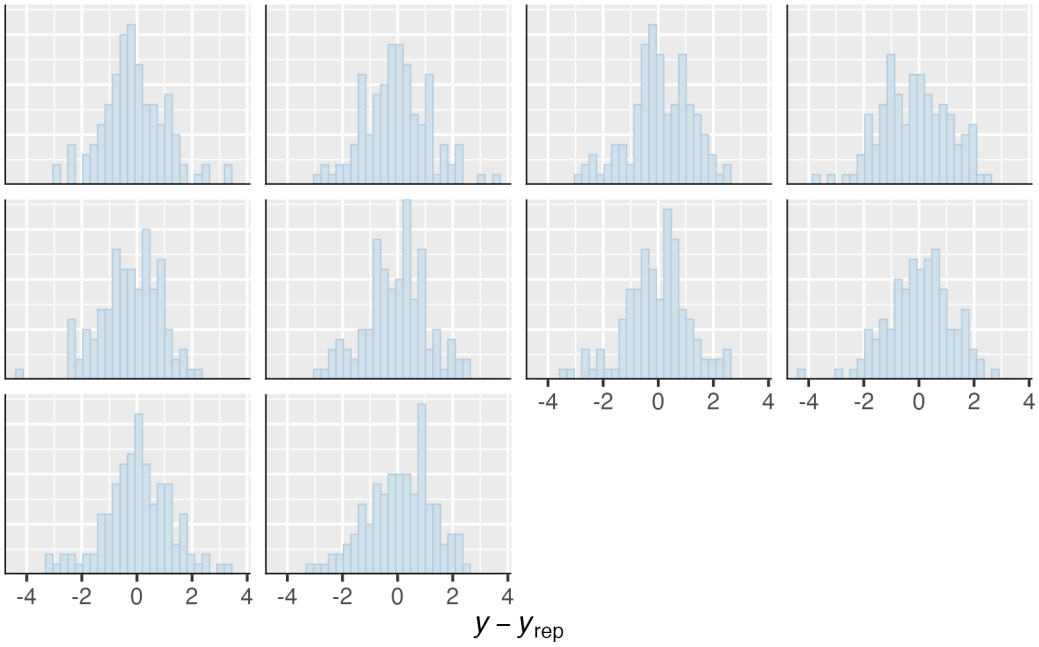


```
=== PPC: Density Overlay (Outcome: smfq_total_s) ===
```

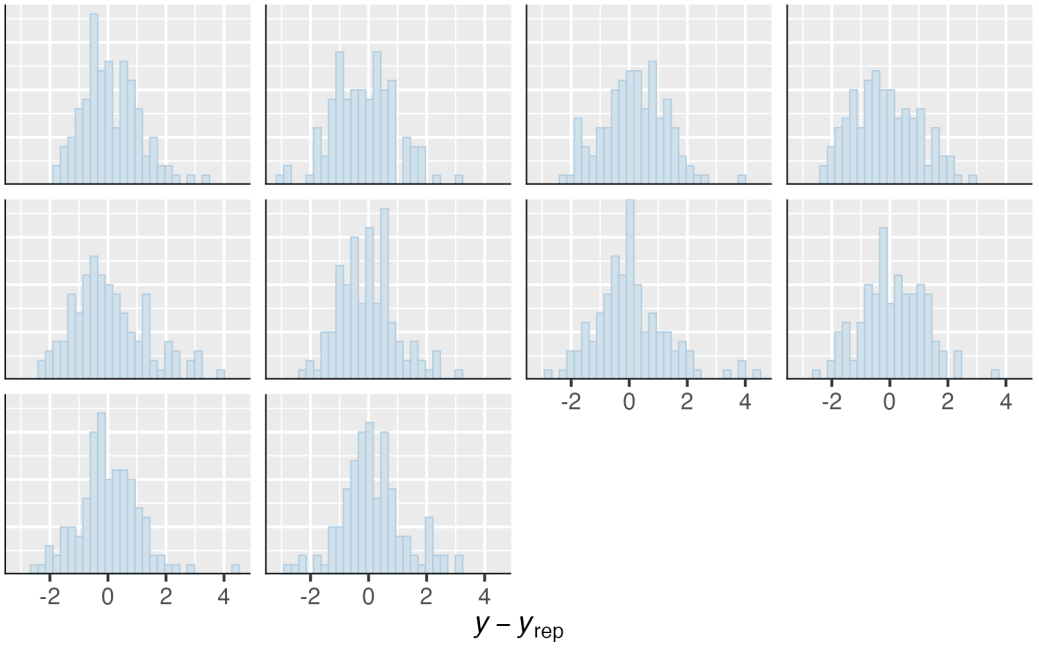



```
=== PPC: Error Histogram (Mediator) ===

Errors should be approximately normally distributed around 0.
```

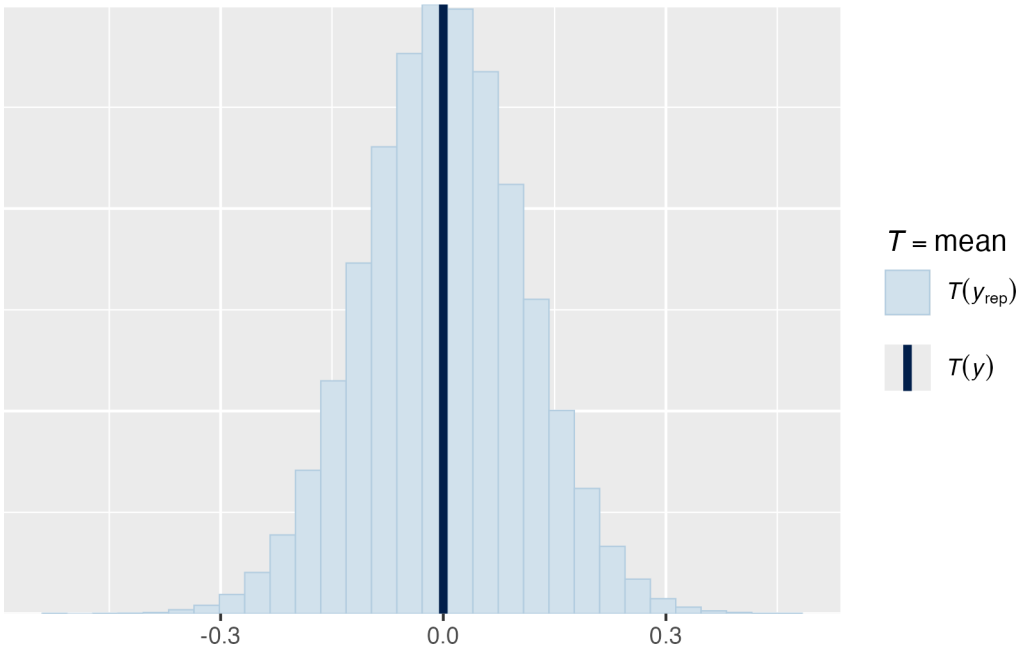



```
=== PPC: Error Histogram (Outcome) ===
```

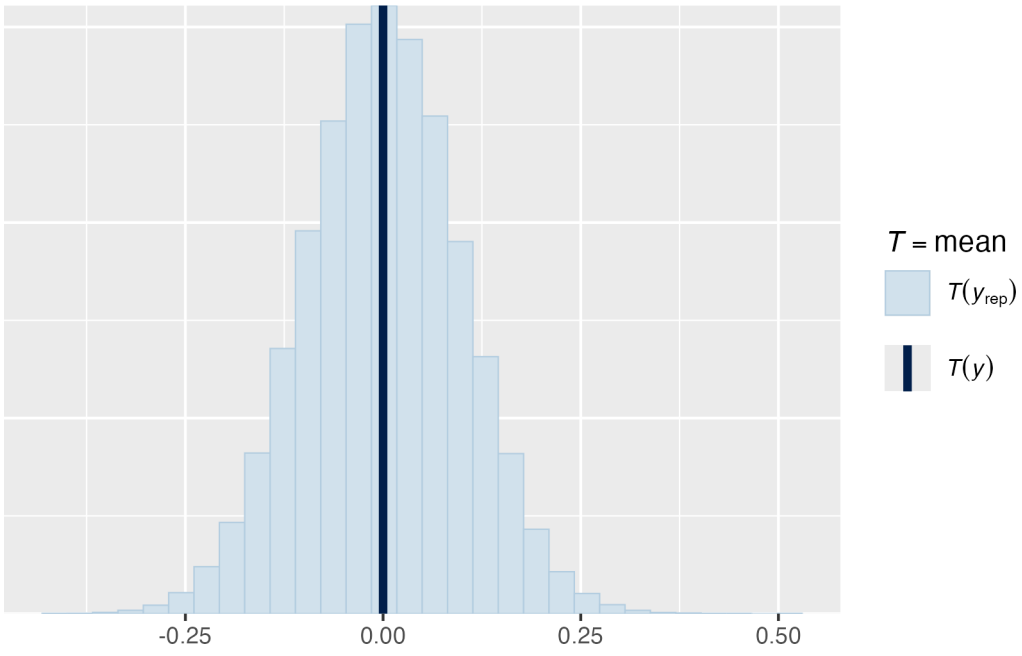



```
=== PPC: Mean (Mediator) ===
```

```
Observed mean (dark line) should fall within the distribution of replicated means.
```

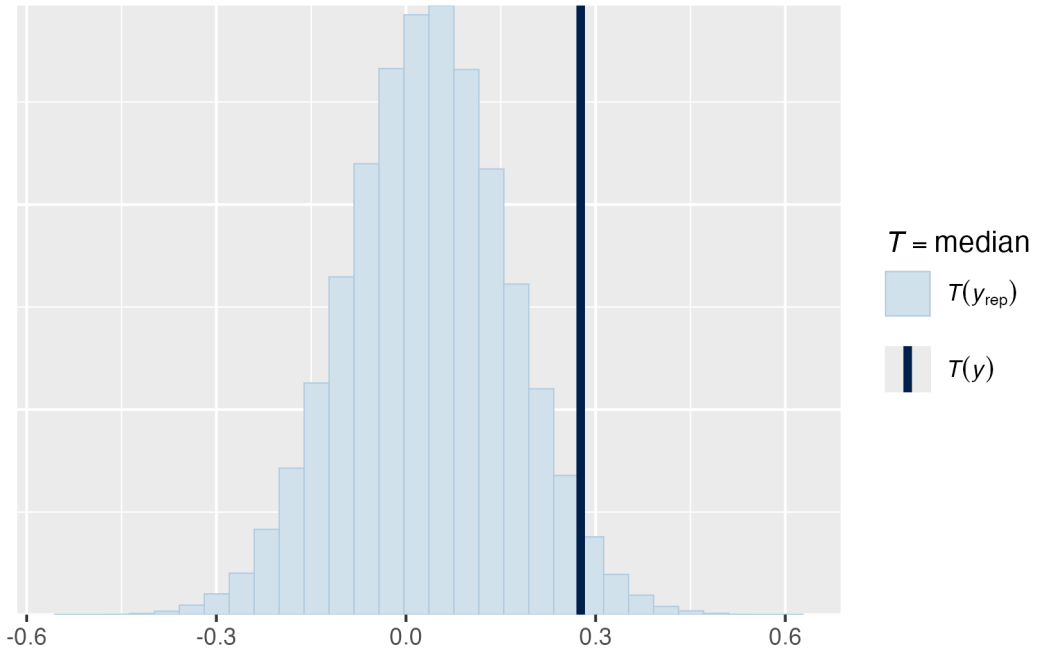



```
=== PPC: Mean (Outcome) ===
```

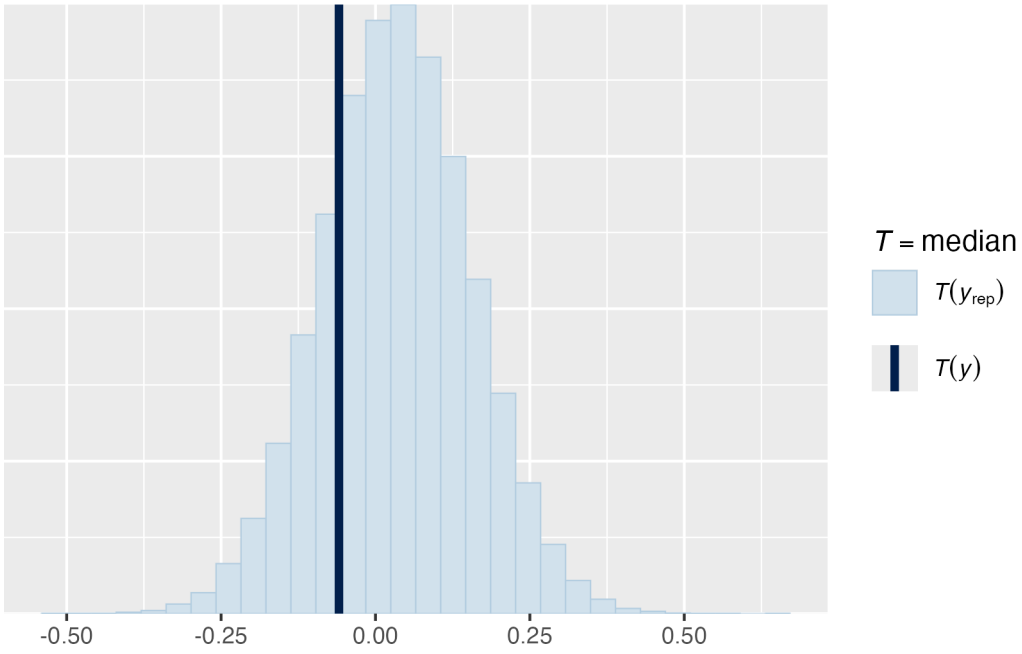


```
=== PPC: Median (Mediator) ===

Observed mean (dark line) should fall within the distribution of replicated means.
```

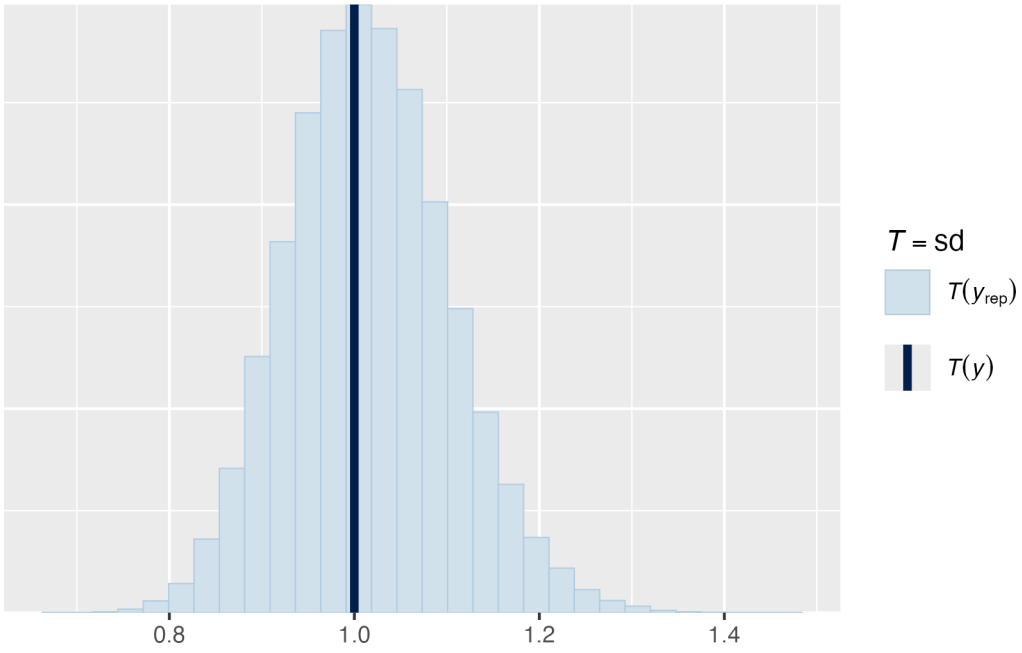



```
=== PPC: Median (Outcome) ===
```

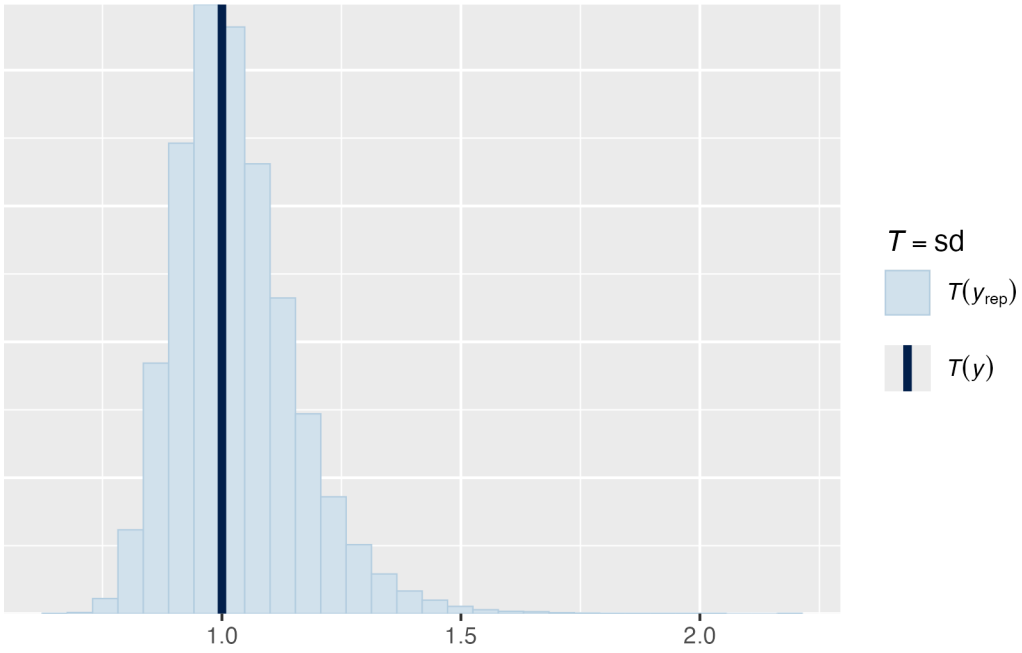



```
=== PPC: SD (Mediator) ===

Observed SD (dark line) should fall within the distribution of replicated SDs.
```

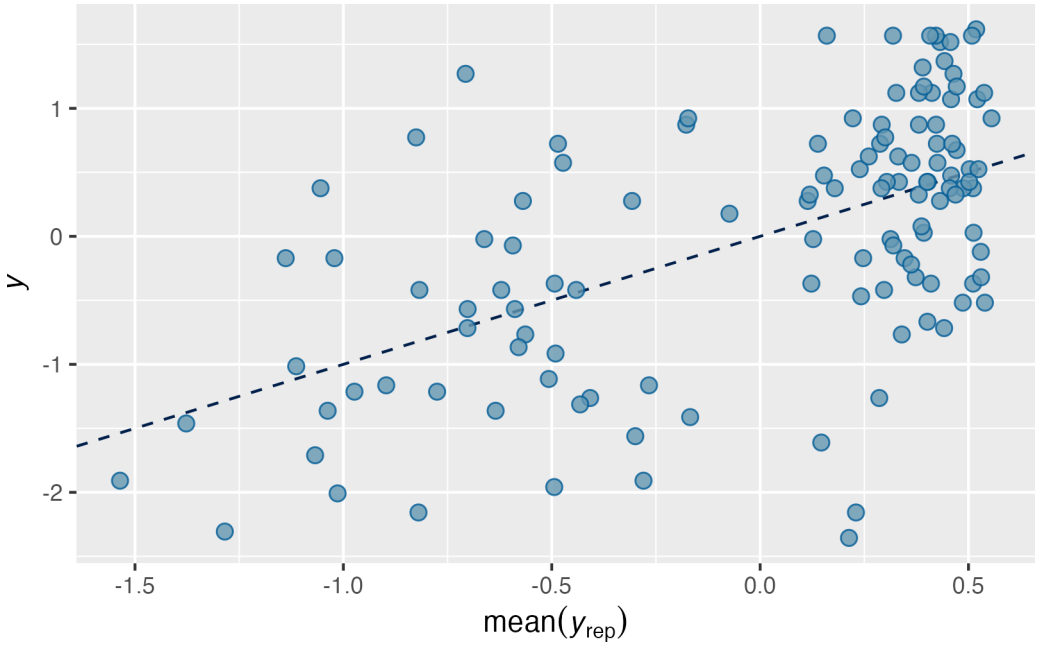


```
=== PPC: SD (Outcome) ===
```

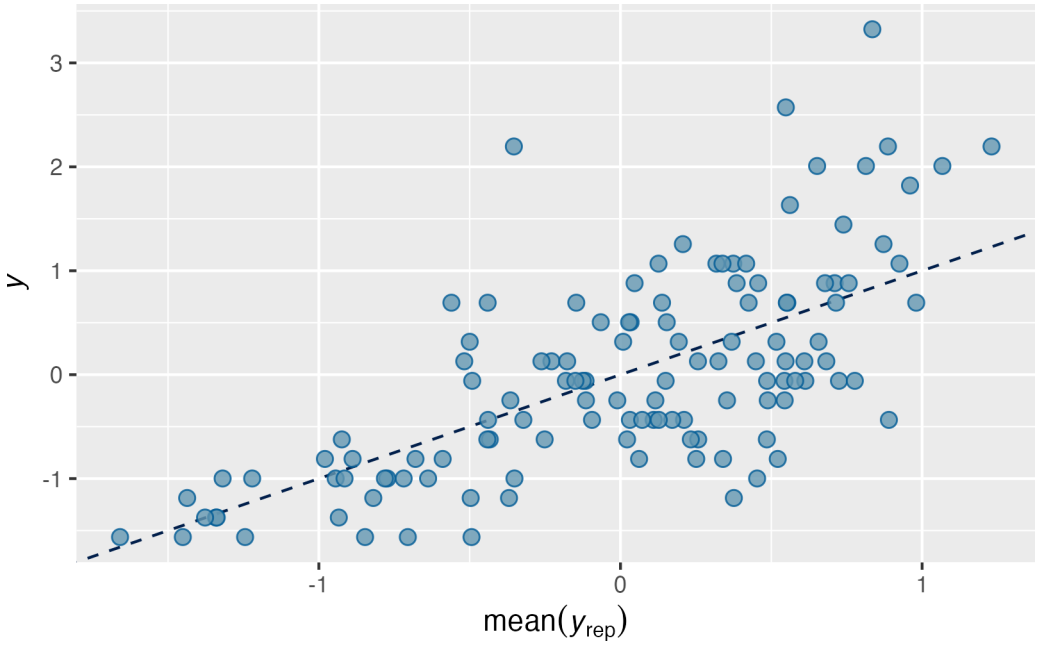



```
=== PPC: Scatter Average (Mediator) ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.
```

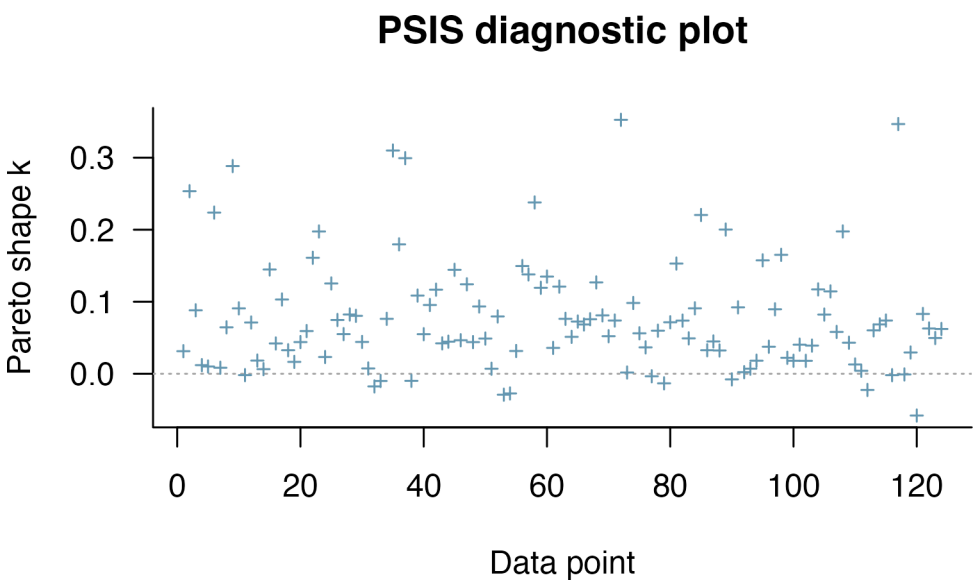



```
=== PPC: Scatter Average (Outcome) ===
```



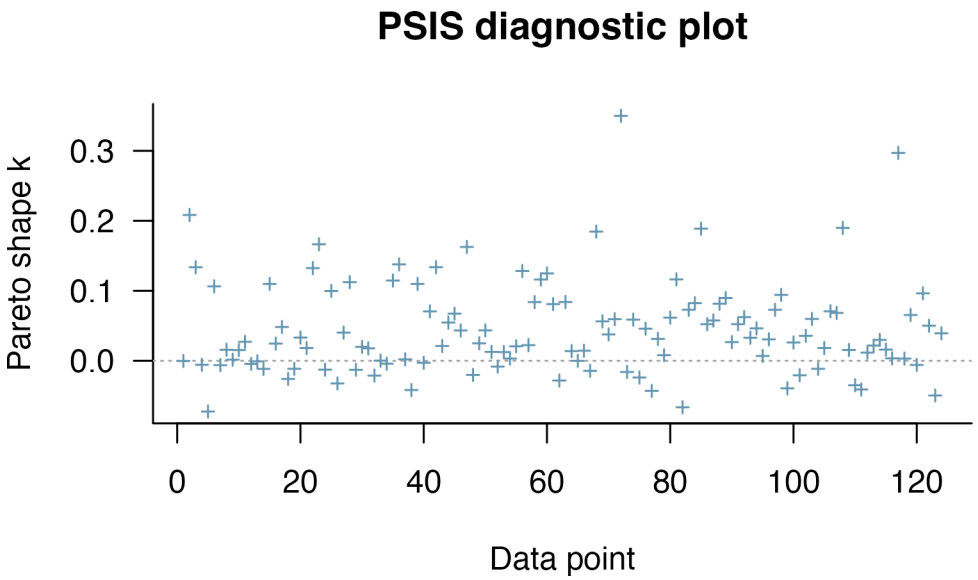

```
=== Pareto-k: Joint model (both equations combined) ===
```

```
All Pareto k estimates are good (k < 0.7).
```




```
=== Pareto-k: Mediator equation only ===
```

```
All Pareto k estimates are good (k < 0.7).
```



```
=== Pareto-k: Outcome equation only ===


All Pareto k estimates are good (k < 0.7).
```

